ANALYSIS OF MALARIA INCIDENCE IN NIGERIA (2000–2024)

A Descriptive Data Analysis Using Microsoft Excel

INTRODUCTION

Malaria remains an important public health concern in Nigeria. Monitoring malaria incidence over time helps us understand changes in the burden of the disease and assess progress in malaria control.

This project examines malaria incidence in Nigeria from 2000 to 2024. The analysis focuses on identifying trends, fluctuations, the highest and lowest incidence rates, and the overall percentage change during the study period.

DATA SOURCE

The data used for this analysis was obtained from Our World in Data (OWID). The dataset contains malaria incidence records for Nigeria covering the period 2000–2024.

The indicator analyzed was malaria incidence per 1,000 people at risk.

Source: Our World in Data — https://ourworldindata.org/malaria

METHODOLOGY

Microsoft Excel was used to clean, organize, analyze and visualize the malaria incidence data.

The following steps were carried out:

1. **Data filtering:** The dataset was filtered to display records for Nigeria only.
2. **Data sorting:** The records were sorted by malaria incidence and year to identify patterns and organize the data chronologically.
3. **Descriptive analysis:** The average, minimum, maximum and range of malaria incidence were calculated.

4. **Percentage change:** The percentage change in incidence between 2000 and 2024 was calculated.
5. **Data visualization:** A line chart was created to show changes in malaria incidence over the study period.

RESULTS AND FINDINGS.

Indicator	Result
Incidence in 2000	401.79
Incidence in 2024	294.25
Highest incidence (2008)	402.01
Lowest incidence (2015)	283.81
Average incidence	339.18
Range	118.20
Percentage decrease	26.77 %

INTERPRETATION OF FINDINGS

The malaria incidence rate in Nigeria generally declined between 2000 and 2024, although the decline was not consistent throughout the period. Incidence was 401.79 per 1,000 people at risk in 2000 and decreased to 294.25 in 2024, representing an overall reduction of approximately 26.77%.

The highest incidence was recorded in 2008 at 402.01, while the lowest was recorded in 2015 at 283.81. This indicates that malaria incidence fluctuated over the years but showed an overall downward trend.

The decrease suggests a positive development in malaria control in Nigeria. However, the data alone cannot establish the factors responsible for this reduction.

DATA VISUALIZATION.

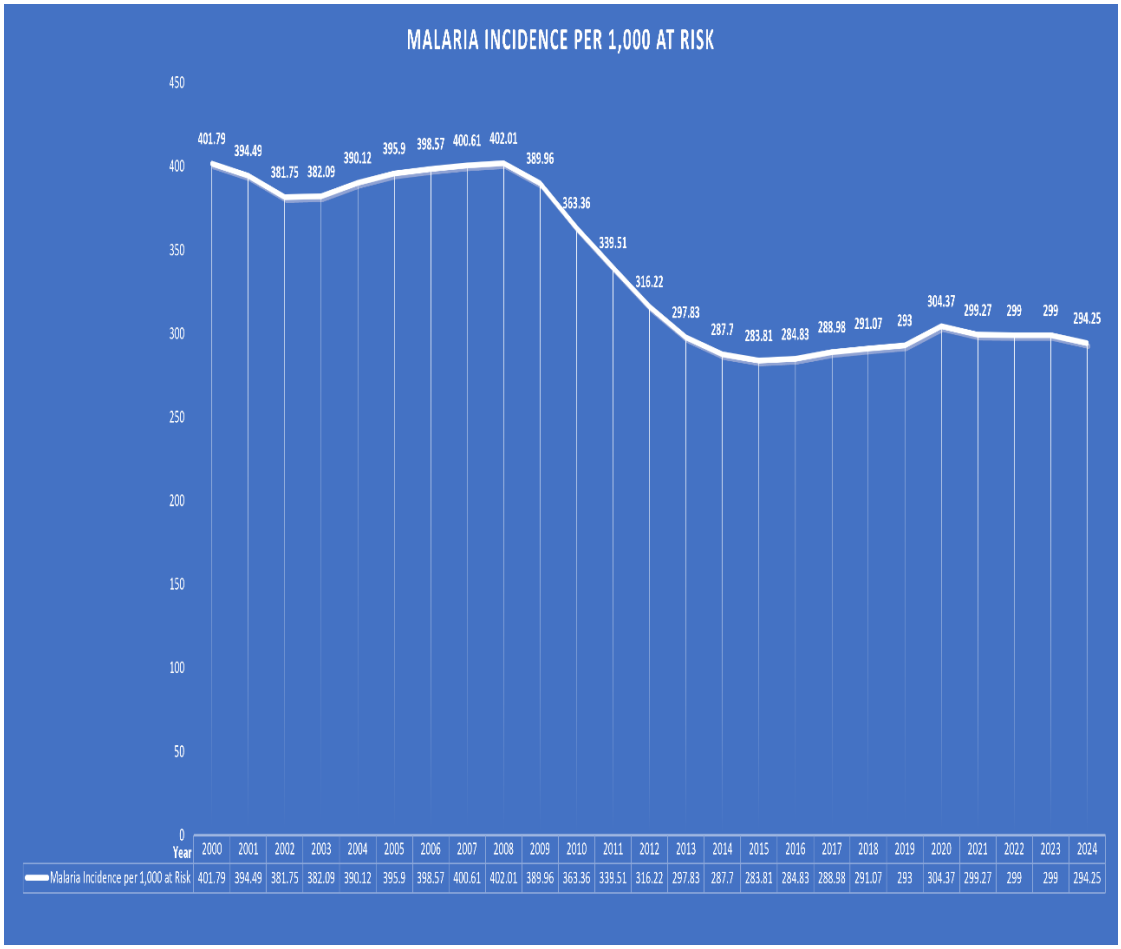


Figure 1: Malaria incidence rate in Nigeria (2000–2024)

CONCLUSION

The analysis revealed an overall decline in malaria incidence in Nigeria between 2000 and 2024, despite fluctuations during the period. The findings suggest progress in malaria control, although further investigation would be needed to understand the factors responsible for the observed changes.